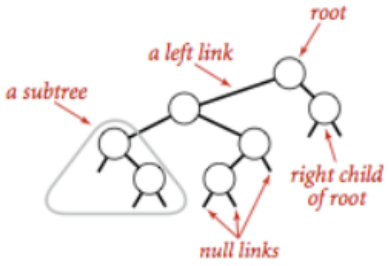
Java 9 - BST - Detyra për ushtrime

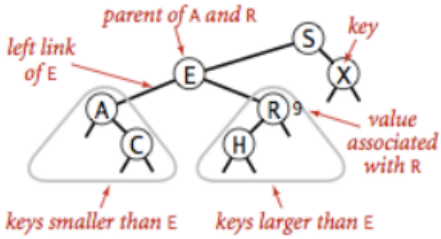
Lënda: Hyrje në Algoritme

Binary Search Tree (BST):

- Është Data Structure
- Implementon Symbol Tables
- Përdoret për kërkim (searching)



Anatomy of a binary tree



Anatomy of a binary search tree

Detyra 1: Nga vargu i dhënë **A = {24,22,94,19,62,39,88,32,40,41,27}**:

a. Të konstrukohet pema me **BST**

b. Të fshihet elementi me vlerë **62**

c. Të shtohet elementi me vlerë **90**

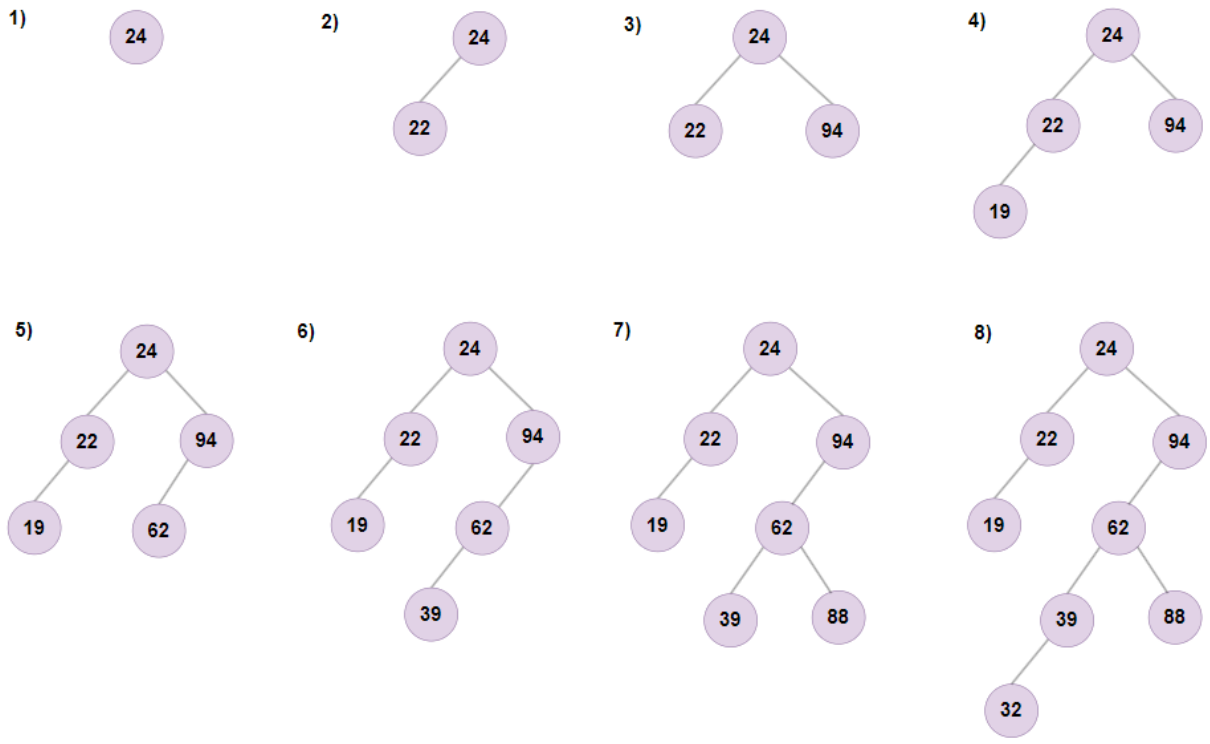
d. Të vizitohet pema me anë të: **InOrder**, **PostOrder** dhe **PreOrder**

e. Gjeni **floor(68) ceiling(68) get(68) get(40)**

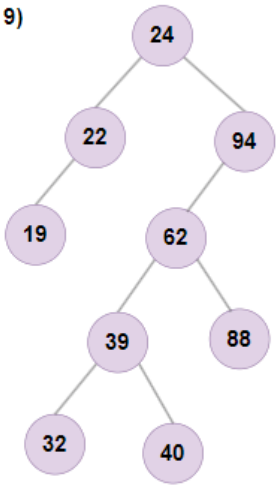
a)

Rregulla:

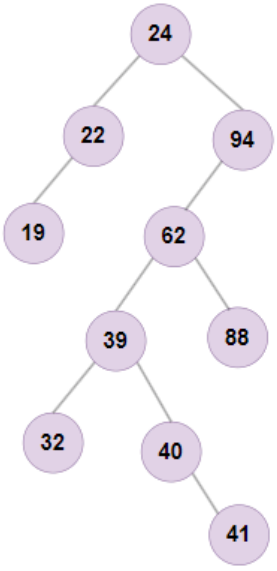
- Cdo **node (nyje)** në anën e majtë e ka fëmijën më të vogël, ndërsa në anën e djathtë më të madh.
- Cdo herë kur shtojmë një element, duhet të krahasohet prej root, për ta caktuar në cilën anë duhet të vendoset.



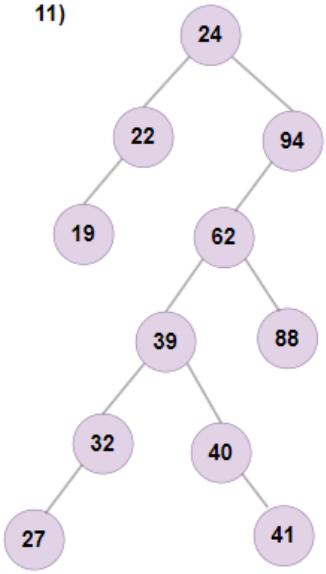
9)



10)



11)

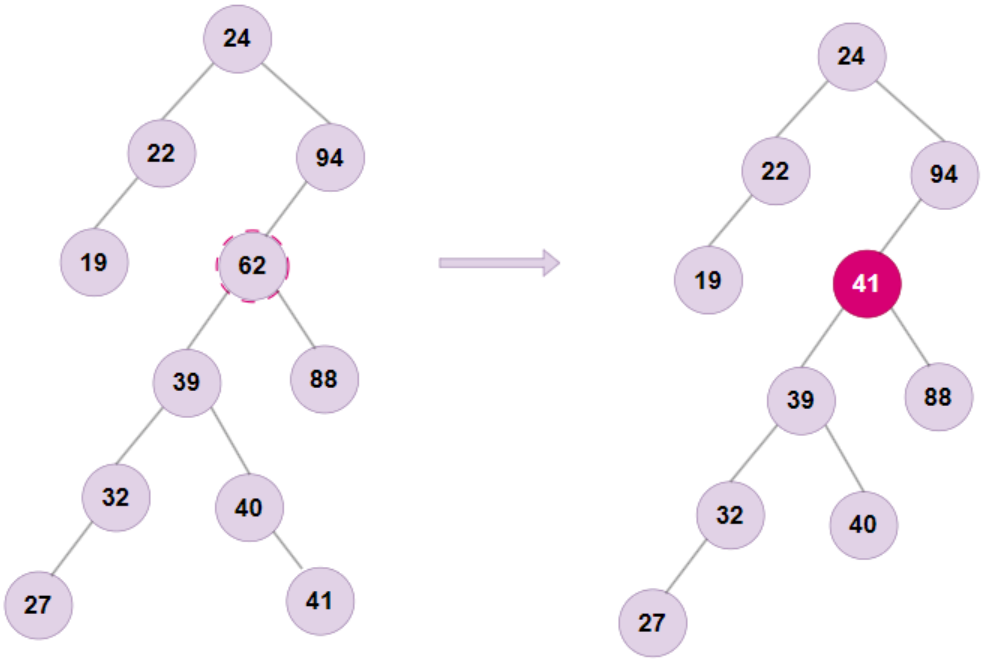


b) I kemi 2 mundësi :

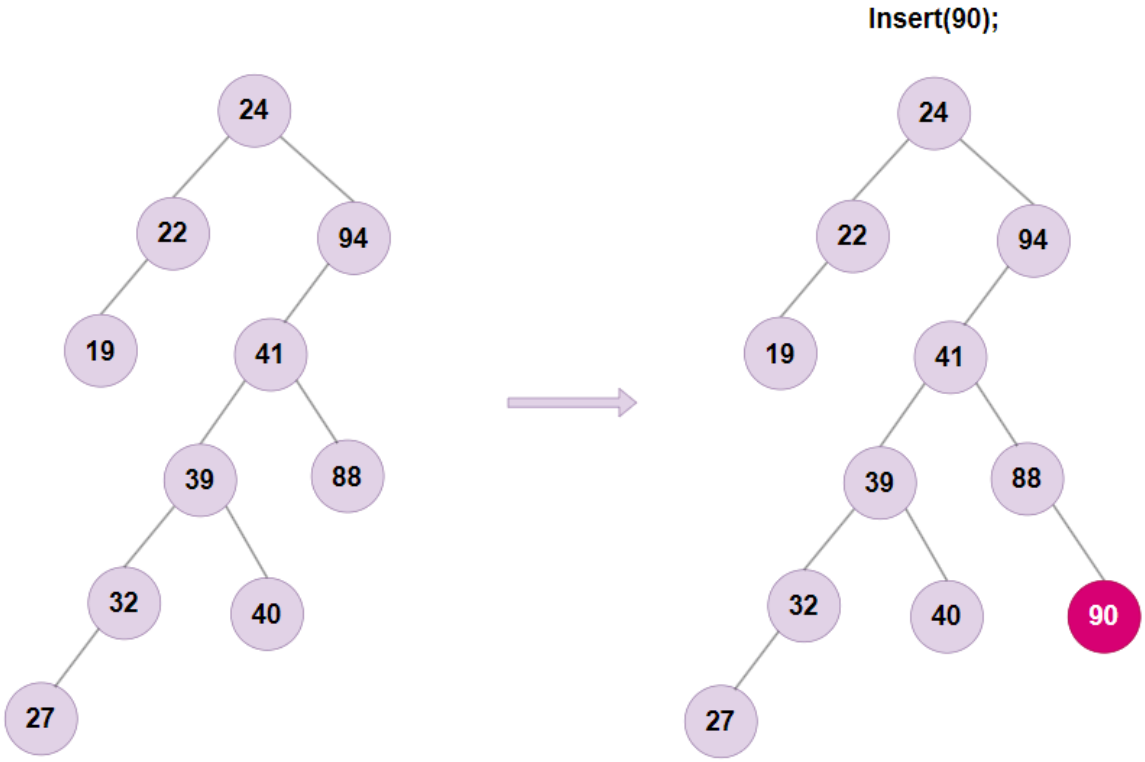
1. *Elementin e zv. me nyjen më të madhe nga dega e majtë e tij* **(në këtë rast me 41)**
2. *Elementin e zv. me nyjen më të vogël nga dega e djathtë e tij* **(në këtë rast me 88)**

Kur fshijmë një element, tentojmë me shkurtu drurin

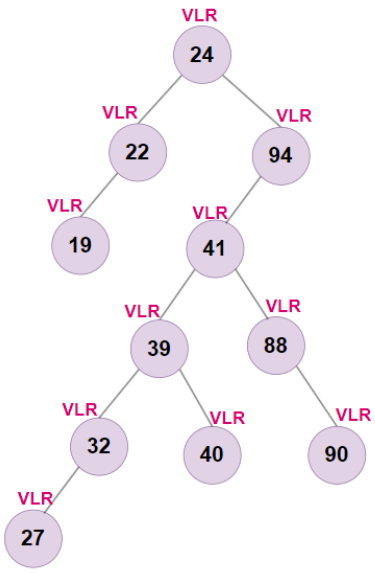
1) delete 62



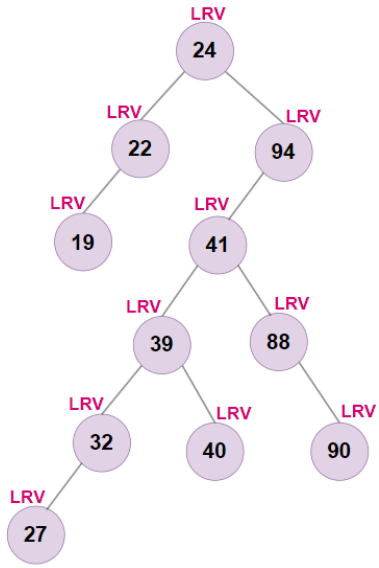
c)



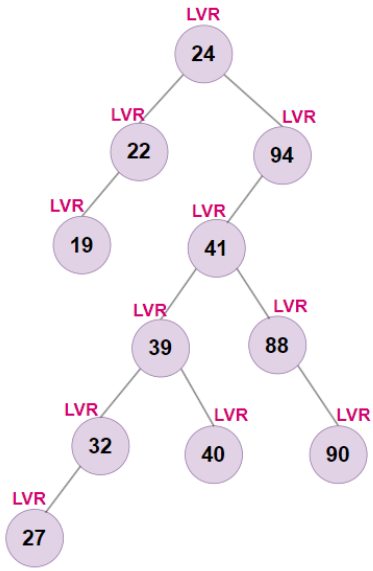
d)



PreOrder: 24,22,19,94,41,39,32,27,40,88,90

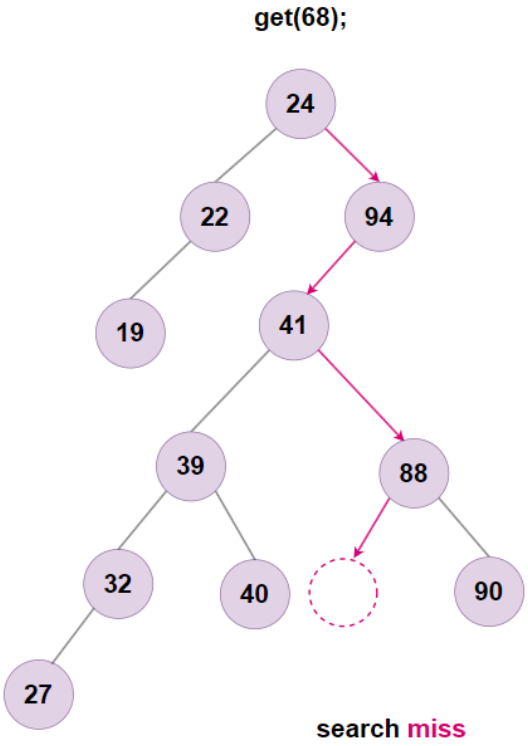


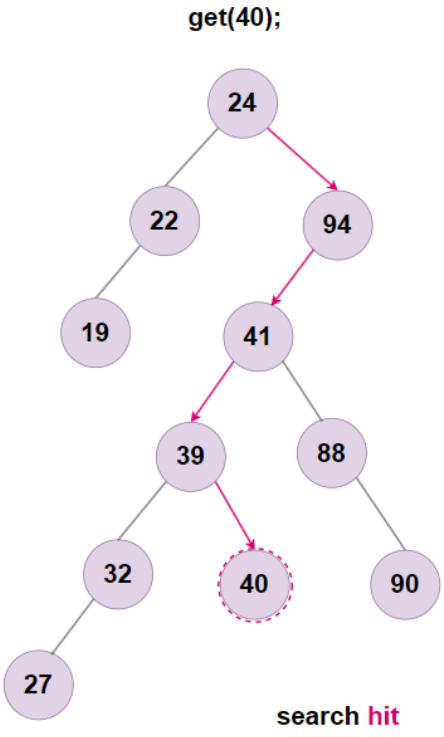
PostOrder: 19,22,27,32,40,39,90,88,41,94,24



InOrder: 19,22,24,27,32,39,40,41,88,90,94

e)





floor(68); // -> 41

ceiling(68); // -> 88

Detyra 2: Nga vargu i dhënë **A = { M, C, Q, A, H, L, P}**:

a. Të konstrukohet pema me **BST**

b. Të fshihet elementi me vlerë **M**

c. Të shtohet elementi me vlerë **J**

d. Të vizitohet pema me anë të: **PostOrder** dhe **PreOrder**

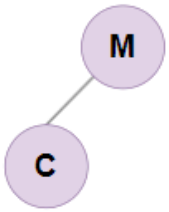
e. Gjeni **floor('B') ceiling('B') get('L') get('T')**

a)

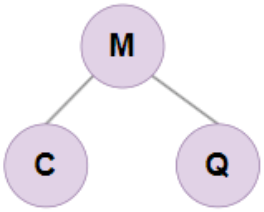
1)



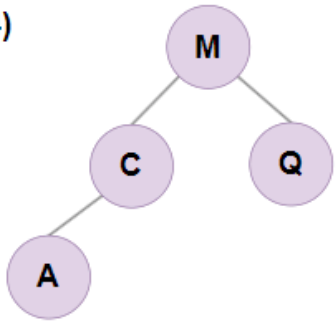
2)



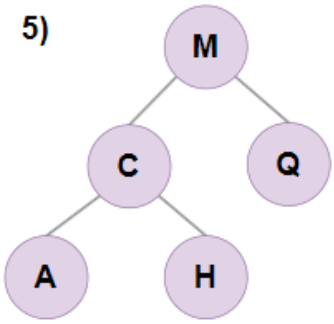
3)



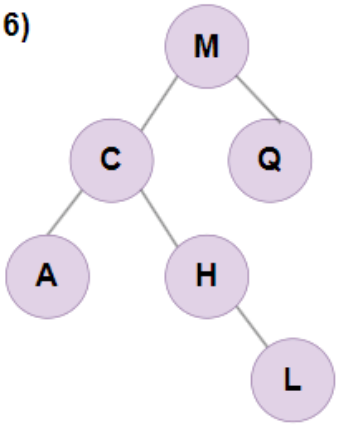
4)



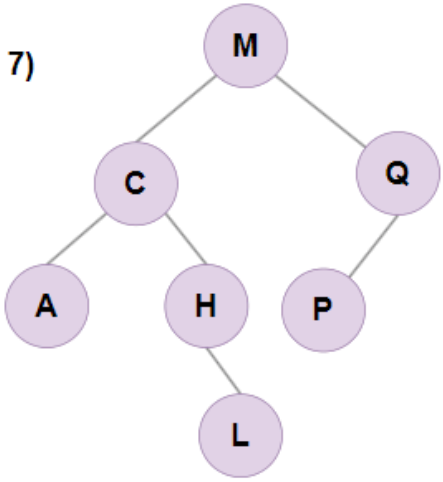
5)



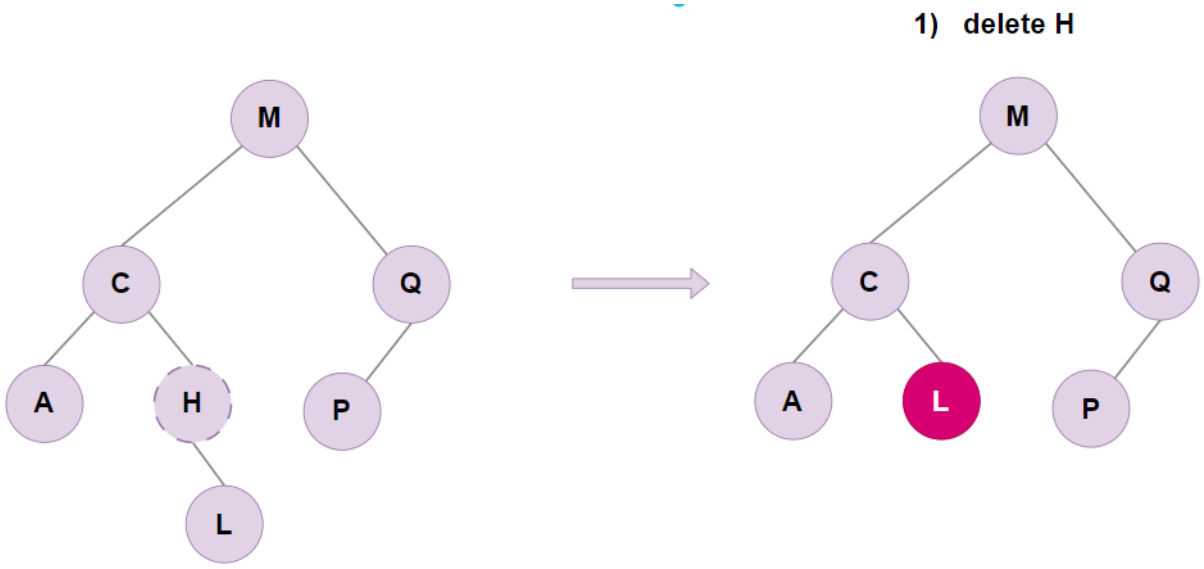
6)



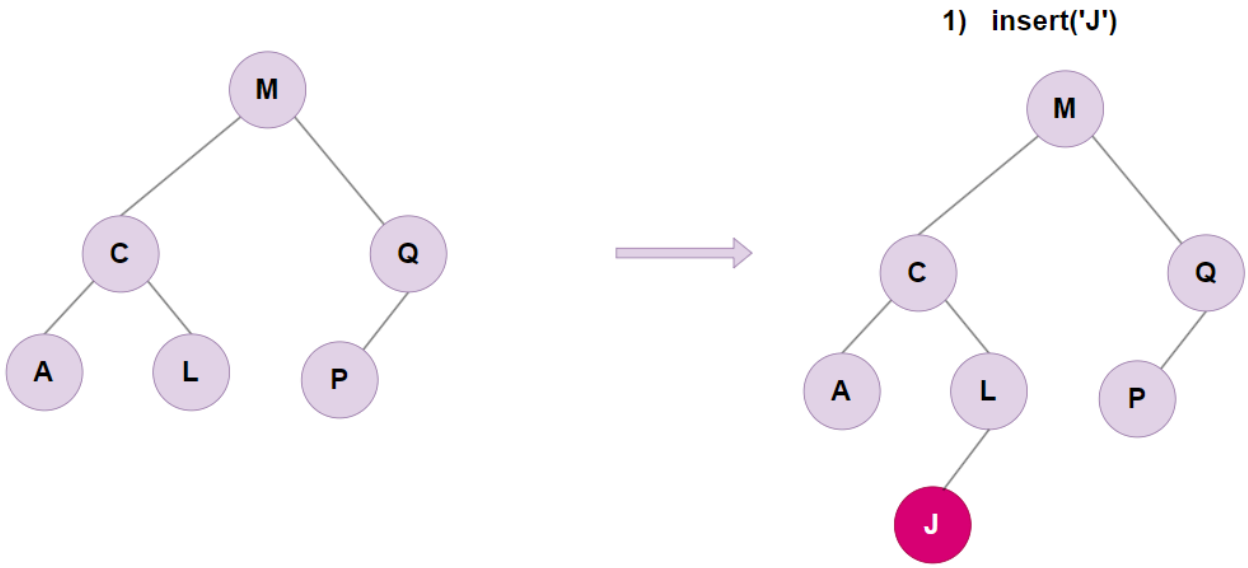
7)



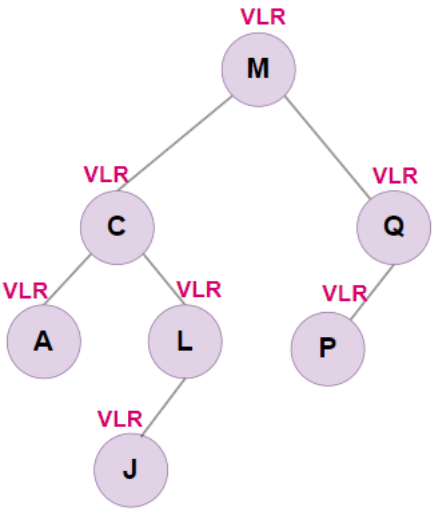
b)



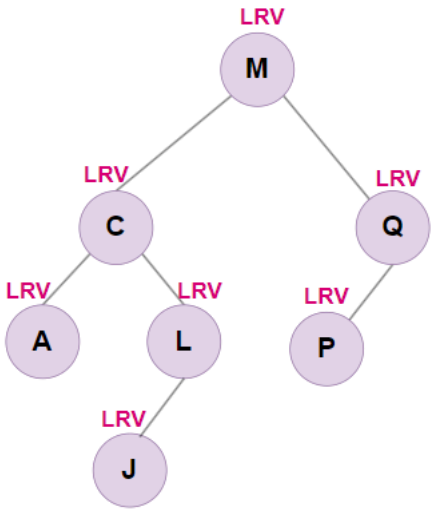
c)



d)



PREORDER: M, C, A, L, J, Q, P

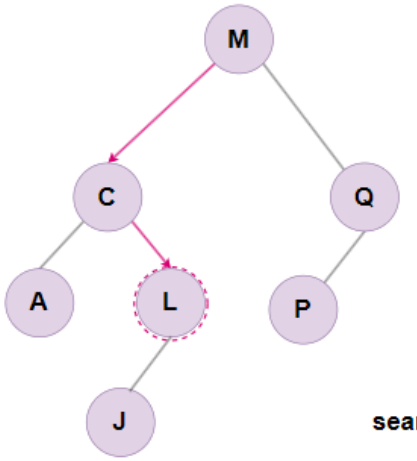


POSTORDER: A, J, L, C, P, Q, M

e)

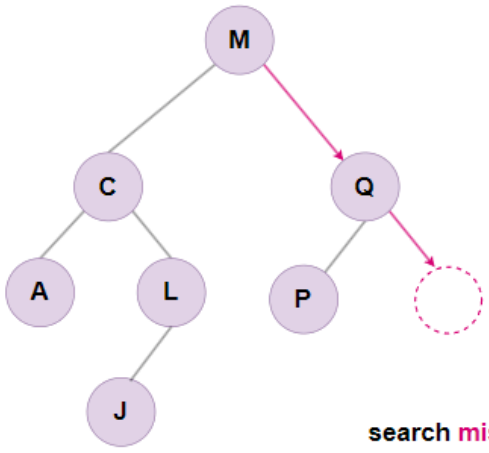
floor(B); // -> A
ceiling(B); // -> C

get('L');



search hit

get('T');



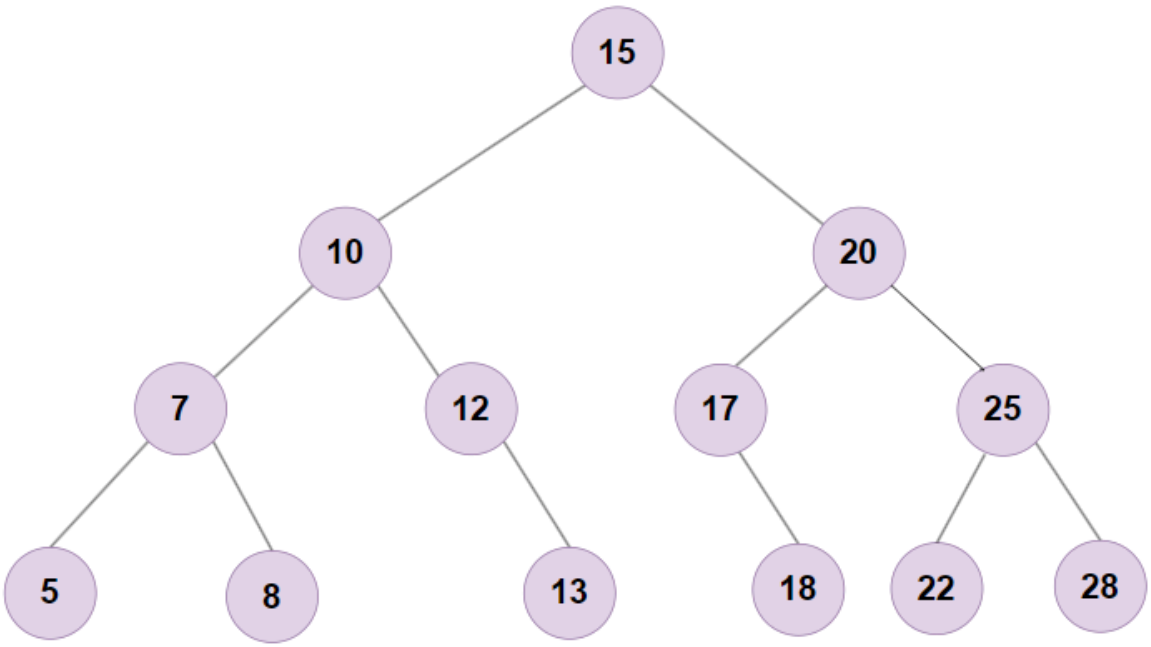
search miss

Detyra 3: Janë dhënë kombinimet e elementeve të një BST në PreOrder dhe PostOrder. Duke u bazuar në to, të ndërtohet trungu me BST.

PreOrder: 15,10,7,5,8,12,13,20,17,18,25,22,28

PostOrder: 5,8,7,13,12,10,18,17,22,28,25,20,15

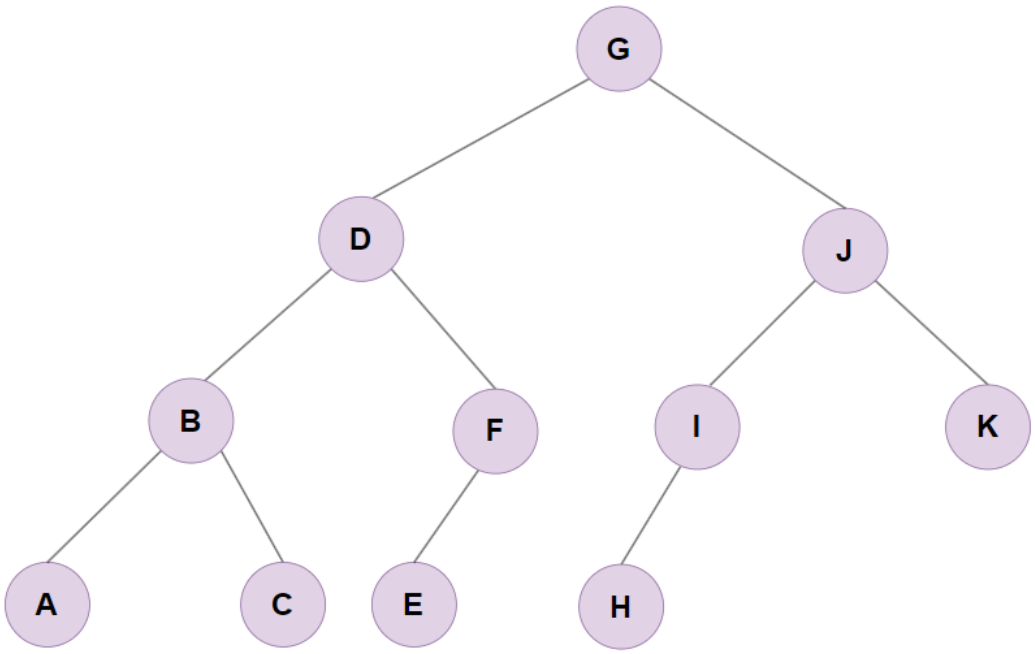
Zgjidhja:



Detyra 4: Është dhënë kombinimi i elementeve të një BST në PostOrder. Duke u bazuar në të, të ndërtohet trungu me BST.

PostOrder: A, C, B, E, F, D, H, I, K, J, G

Zgjidhja:



Links për visualizim të BST :

https://yongdanielliang.github.io/animation/web/BST.html (for numbers)

https://www.cs.usfca.edu/~galles/visualization/BST.html (for letters)